

Design and Implementation of Secure Network Infrastructure using Defense-in-Depth Framework

Ali Wael Ali Soliman, Ali Mohamed Ahmed, Ahmed Sayed El-Arabi, Nermin Awsy Khalil,
Miral Alaa Hamed, Mariam Mohsen, Esraa Mohamed

*Egyptian Chinese College at Suez Canal University
Telecommunications and Network Engineering Department*

Abstract

The Secure Cloud project presents a secure hierarchical enterprise network architecture designed to replace traditional flat-network environments with a segmented and defense-oriented infrastructure. The report demonstrates the implementation of VLAN segmentation, Inter-VLAN routing, OSPF authentication, IPsec VPN encryption, port security, and management plane hardening using Cisco Packet Tracer technologies. The proposed architecture follows the Defense-in-Depth strategy by deploying multiple security layers across Layer 2, Layer 3, and management services. The objective is to minimize lateral movement, mitigate routing manipulation attacks, secure remote connectivity, and provide centralized monitoring and logging capabilities. Verification procedures, threat analysis, and security comparisons are included to validate operational integrity and resilience against common enterprise network attacks.

I. Introduction

Modern organizations require secure, scalable, and manageable infrastructures capable of supporting cloud services, internal communication, and secure remote access. Traditional flat networks expose all devices to the same broadcast domain, increasing the risk of unauthorized access, broadcast storms, and unrestricted lateral movement. The architecture transforms this insecure structure into a hierarchical segmented model where traffic is logically separated and controlled through secure routing boundaries.

The implemented architecture adopts a Defense-in-Depth model. Instead of relying on a single firewall or isolated security mechanism, multiple layers of protection are deployed simultaneously. VLAN segmentation limits exposure, routing authentication protects topology integrity, VPN encryption secures data in transit, and access control policies restrict unauthorized communications. Management hardening mechanisms further protect routers and switches from brute-force attacks and unauthorized administrative access.

II. VLAN Segmentation and Network Isolation

Security Threat:

Flat networks expose all hosts to broadcast traffic and unauthorized lateral movement. Attackers can perform ARP spoofing, packet sniffing, and VLAN hopping attacks.

Mitigation Strategy:

VLANs isolate departments into independent broadcast domains. Devices from different VLANs require Layer 3 routing for communication, allowing administrators to inspect and filter traffic using ACLs and firewalls.

Technical Implementation:

```
Switch(config)#vlan 10
Switch(config-vlan)#name HR
Switch(config)#vlan 20
Switch(config-vlan)#name IT
Switch(config)#interface fa0/1
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 10
Switch(config)#interface fa0/24
Switch(config-if)#switchport trunk encapsulation dot1q
Switch(config-if)#switchport mode trunk
Switch(config-if)#switchport trunk allowed vlan 10,20
```

Verification & Testing:

```
R1# show running-config
R1# show vlan brief
R1# show interfaces trunk
R1# show ip route
R1# show ip ospf neighbor
R1# show port-security interface fa0/2
R1# show crypto isakmp sa
R1# show ssh
```

The verification phase confirmed successful routing convergence, secure tunnel establishment, VLAN isolation, and restricted administrative access.

III. Inter-VLAN Routing Security

Security Threat:

Without secure routing boundaries, unauthorized hosts may access sensitive resources across VLANs. Misconfigured routing enables lateral movement attacks.

Mitigation Strategy:

Router-on-a-Stick and Layer 3 switching centralize traffic inspection. ACLs and routing policies restrict unauthorized communication between departments.

Technical Implementation:

```
R1(config)#interface fa0/0.10
R1(config-subif)#encapsulation dot1Q 10
R1(config-subif)#ip address 192.168.110.1 255.255.255.0
R1(config)#interface fa0/0.20
R1(config-subif)#encapsulation dot1Q 20
R1(config-subif)#ip address 192.168.120.1 255.255.255.0
R1(config)#access-list 110 deny tcp 192.168.110.0 0.0.0.255 host 192.168.120.10 eq 80
R1(config)#access-list 110 permit ip any any
```

Verification & Testing:

```
R1# show running-config
R1# show vlan brief
R1# show interfaces trunk
R1# show ip route
R1# show ip ospf neighbor
R1# show port-security interface fa0/2
R1# show crypto isakmp sa
R1# show ssh
```

The verification phase confirmed successful routing convergence, secure tunnel establishment, VLAN isolation, and restricted administrative access.

IV. OSPF Authentication and Routing Integrity

Security Threat:

Attackers may inject fake routing updates or deploy rogue routers to manipulate traffic paths, causing route poisoning or Man-in-the-Middle attacks.

Mitigation Strategy:

OSPF MD5 authentication ensures that only trusted routers sharing the correct authentication key can exchange routing information.

Technical Implementation:

```
R1(config)#router ospf 1
```

```
R1(config-router)#network 192.168.1.0 0.0.0.3 area 0
R1(config)#interface serial0/0/0
R1(config-if)#ip ospf message-digest-key 1 md5 ajex2026
R1(config-if)#ip ospf authentication message-digest
```

Verification & Testing:

```
R1# show running-config
R1# show vlan brief
R1# show interfaces trunk
R1# show ip route
R1# show ip ospf neighbor
R1# show port-security interface fa0/2
R1# show crypto isakmp sa
R1# show ssh
```

The verification phase confirmed successful routing convergence, secure tunnel establishment, VLAN isolation, and restricted administrative access.

V. Layer 2 Protection and Port Security

Security Threat:

MAC flooding and MAC spoofing attacks may force switches to broadcast traffic, allowing attackers to sniff network communications.

Mitigation Strategy:

Port Security limits the number of MAC addresses per interface and blocks unauthorized devices.

Technical Implementation:

```
Switch(config)#interface range fa0/2-24
Switch(config-if-range)#switchport mode access
Switch(config-if-range)#switchport port-security
Switch(config-if-range)#switchport port-security maximum 1
Switch(config-if-range)#switchport port-security mac-address sticky
Switch(config-if-range)#switchport port-security violation restrict
```

Verification & Testing:

```
R1# show running-config
R1# show vlan brief
R1# show interfaces trunk
R1# show ip route
R1# show ip ospf neighbor
R1# show port-security interface fa0/2
R1# show crypto isakmp sa
R1# show ssh
```

The verification phase confirmed successful routing convergence, secure tunnel establishment, VLAN isolation, and restricted administrative access.

VI. Site-to-Site IPsec VPN

Security Threat:

Data transmitted between branches over public networks can be intercepted through packet sniffing or session hijacking.

Mitigation Strategy:

IPsec VPN creates encrypted tunnels using AES encryption and SHA hashing to ensure confidentiality and integrity.

Technical Implementation:

```
R1(config)#crypto isakmp policy 10
R1(config-isakmp)#encryption aes
R1(config-isakmp)#hash sha
R1(config-isakmp)#authentication pre-share
R1(config)#crypto isakmp key agex address 192.168.3.1
R1(config)#crypto ipsec transform-set VPN_SET esp-aes esp-sha-hmac
R1(config)#crypto map VPN_MAP 10 ipsec-isakmp
R1(config-crypto-map)#set peer 192.168.3.1
R1(config-crypto-map)#set transform-set VPN_SET
```

Verification & Testing:

```
R1# show running-config
R1# show vlan brief
R1# show interfaces trunk
R1# show ip route
R1# show ip ospf neighbor
R1# show port-security interface fa0/2
R1# show crypto isakmp sa
R1# show ssh
```

The verification phase confirmed successful routing convergence, secure tunnel establishment, VLAN isolation, and restricted administrative access.

VII. Management Plane Hardening

Security Threat:

Routers and switches are vulnerable to brute-force login attempts, Telnet sniffing, and unauthorized remote management access.

Mitigation Strategy:

SSH replaces insecure Telnet sessions. Local authentication databases, encrypted passwords, login restrictions, and banners strengthen management security.

Technical Implementation:

```
R1(config)#hostname R1
```

```
R1(config)#enable secret agex Admin
R1(config)#username admin privilege 15 secret StrongPass2026
R1(config)#ip domain-name agex.local
R1(config)#crypto key generate rsa
R1(config)#ip ssh version 2
R1(config)#line vty 0 4
R1(config-line)#login local
R1(config-line)#transport input ssh
R1(config)#login block-for 120 attempts 3 within 60
```

Verification & Testing:

```
R1# show running-config
R1# show vlan brief
R1# show interfaces trunk
R1# show ip route
R1# show ip ospf neighbor
R1# show port-security interface fa0/2
R1# show crypto isakmp sa
R1# show ssh
```

The verification phase confirmed successful routing convergence, secure tunnel establishment, VLAN isolation, and restricted administrative access.

VIII. Security Comparison Tables

Feature	Static Routing	Dynamic Routing	Security Impact
Configuration	Manual	Automatic	Dynamic routing requires authentication
Scalability	Low	High	Dynamic protocols increase attack surface
Updates	Administrator controlled	Automatic exchanges	Untrusted updates may cause route injection
Overhead	Low	Medium/High	OSPF authentication mitigates risks

Threat	Mitigation Technology	Security Benefit
MAC Spoofing	Port Security	Unauthorized devices are blocked
Route Injection	OSPF MD5 Authentication	Trusted routing only
Packet Sniffing	IPsec VPN	Encrypted communication
Brute Force	SSH Hardening	Encrypted and restricted access
Lateral Movement	ACLs + VLANs	Traffic segmentation

IX. Conclusion

The project demonstrates how modern enterprise networks can transition from insecure flat architectures into secure hierarchical infrastructures using Defense-in-Depth principles. Through VLAN segmentation, authenticated routing, encrypted VPN tunnels, ACL enforcement, and hardened management access, the network significantly reduces attack exposure while improving scalability and operational control.

The implementation proved that layered security mechanisms are essential for defending against both internal and external threats. Future improvements may include Intrusion Detection Systems (IDS), Network Access Control (NAC), and centralized SIEM monitoring to further strengthen enterprise visibility and threat response capabilities.